Experiment 6:

Code:

```
  GNU nano 9.0
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
int mb;

if(argc != 2)
{
printf("Usage: %s <memory_in_MB>\n", argv[0]);
return 1;
}

mb = atoi(argv[1]);

char *memory = malloc(mb*1024*1024);

if(memory == NULL)
{
printf("Memory allocation failed\n");
return 1;
}

for(int i = 0; i <  mb*1024*1024; i += 4096)
{
memory[i] = 1;
}
printf("%d Mb memory allocated.\n", mb);
printf("Program is running... Press Ctrl+C to exit.\n");

while(1)
{
sleep(1);
}
return 0;
}
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ free -h
               total        used        free      shared
Mem:           3.8Gi       1.5Gi       1.2Gi        87Mi
Swap:          953Mi          0B       953Mi

┌──(kali㉿sourav)-[~]
└─$ ./exp6 800
800 Mb memory allocated.
Program is running... Press Ctrl+C to exit.
```

```
┌──(kali㉿sourav)-[~]
└─$ free -h
               total        used        free      shared
Mem:           3.8Gi       2.3Gi       400Mi        94Mi
Swap:          953Mi          0B       953Mi
```